

STATEMENT OF WORK
National Oceanic and Atmospheric Administration (NOAA)
NOAA Fisheries
{NOAA Fisheries Southeast Fisheries Science Center (SEFSC)}
{Cetacean Passive Acoustic Monitoring}

PURPOSE:

The SEFSC requires purchase of (1) VR100 Transponding Hydrophone with a 50 meter cable for redeploying acoustic releases, (5) VR2AR acoustic releases for mooring retrieval, (16) VR2AR-X extended deployment acoustic releases for mooring retrieval, and (10) VR2AR Redeployment part kits to release devices for retrieval, in order to complete multiple cetacean passive acoustic monitoring research projects, including those focused on Rice's whales and North Atlantic right whales. This equipment is required for remote retrieval of moored oceanographic equipment, specifically moored underwater acoustic recorders. Moored underwater acoustic recorders allow for continuous, long-term recordings of ocean acoustics and marine animal sounds. These recorders are essential to data collection targeted at addressing the mission critical project identified below, focused on improving ocean noise research and on endangered species distribution and presence. In order to be able to retrieve these recorders from deeper waters and to be able to log coded acoustic tag transmissions, combined coded acoustic receivers / acoustic releases are needed to accompany the recorders on the moorings.

SCOPE OR MISSION:

The SEFSC has been deploying autonomous moored passive acoustic recording units and associated acoustic releases with coded acoustic tag transmission receivers in the habitat of the Rice's whales and North Atlantic Right whales to record whale vocalizations and ocean soundscapes and the presence of tagged fish species. This is a mission critical project focused on improving ocean noise research on endangered species distribution and presence.

BACKGROUND:

The NOAA NMFS uses moored underwater acoustic recorders to record ocean noise and marine animal vocalizations for the purposes of monitoring their distribution and abundance, while gaining a better understanding of the overall soundscape. To recover these moorings, acoustic releases are required that release ballast and allow the recorders and release to float to the surface. Additionally, moorings with combined acoustic releases / coded acoustic receivers for receiving transmissions from acoustic tags allow monitoring of a variety of fish species that have been tagged by fisheries and university researchers to improve understanding of trophic ecology and ecosystem interactions. The moorings are commonly deployed to the seafloor in moderate to deep waters (15-500m) which require VR100-300 Tracking Acoustic Receive transponder systems capable of communicating with surface vessels on the status of the mooring (upright or tilted, secured or released, etc.) to release and recover moored instruments. The VR100 hydrophones enable communication to the VR2AR or VR2AR-X device on the seabed, facilitating localization and release of the unit. These units must be compatible with existing

units in the SEFSC marine mammal branch's inventory and be capable of communicating with the existing deck-box for remote triggering. For these mooring deployments, VR2AR-69kHz Coded Acoustic Receiver with Acoustic Release units or VR2AR-X-69kHz Coded Acoustic Receiver with Acoustic Release Extended Life units are required to receive coded tag transmission and to recover the moored instruments to obtain the data logged by the recorders. SEFSC requires purchase of (5) VR2AR-69kHz Acoustic Releases, (16) VR2AR-X-69kHz Acoustic Releases (Extended Life), and associated accessories for field research projects in 2026. Required accessories include (1) VR100 Transponding Hydrophone with 50 meter cable as required to effectively communicate with the acoustic moorings upon deployment and recovery across multiple deployment locations, and (10) redeployment part kits that are single use and vital in the release mechanism of the units for retrieval.

TECHNICAL SPECIFICATIONS: The SEFSC requires Vemco VR2AR and VR2AR-X acoustic release and receiver units and associated accessories that are capable of communicating with compatible acoustic command deck units, capable of receiving existing coded tag transmissions, and are compatible with existing needs of SEFSC projects (battery life, deployment depth, etc.). The units must be able to communicate with a Vemco VR100 - 8 Channel DSP Manual Tracking Acoustic Receiver transmitting deck box with corresponding VHTx 69kHz Transponding Hydrophone to arm, and activate the acoustic release to release the moored instruments from their anchor for recovery. The (1) VR100 Transponding Hydrophone with 50 meter cable is required to effectively communicate with the acoustic moorings upon deployment and recovery. The (10) VR2AR lugs and redeployment part kits are made specifically for the release mechanism of the acoustic release

Acoustic Release & Receiver capabilities:

- Minimum 1 year battery life for sites with 6-month deployments; minimum 2 year battery life for sites with 12-month deployments
- Capable of deployment depths up to 500m
- A push-off release mechanism made from robust non-corrosive metal with reliable DC motor
- A safe release load of at least 250 kg
- Transmit frequency of 69 kHz
- Ability to arm and activate the release from surface deck box communications
- Ability to decode and log VEMCO 69 kHz coded tag transmissions
- A minimum of 32 MB non-volatile flash memory for tag transmission log storage
- Ability to communicate with a Vemco VR100 - 8 Channel DSP Manual Tracking Acoustic Receiver transmitting deck box with corresponding VHTx 69kHz Transponding Hydrophone

The units must be compatible with existing VR2AR-69kHz acoustic releases in SEFSC's inventory to ensure data can be accurately compared across sites and hardware can be exchanged among sites.

DELIVERY SCHEDULE AND LOCATION:

Delivered by June 1, 2026 to NOAA Southeast Fisheries Science Center, 75 Virginia Beach Drive, Miami FL 33149

- (1) VR100 Transponding Hydrophone with 50 meter cable
- (5) VR2AR-69kHz Coded Acoustic Receiver with Acoustic Release
- (16) VR2AR-X-69kHz Coded Acoustic Receiver with Acoustic Release (Extended Life)
- (10) VR2AR Redeployment Parts kit